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Describes tests for a type of Doppler radar system referred to as a radar chronograph. Radar chronographs, such as the M90, are used by the U.S. Army for measuring muzzle velocity in field artillery applications. Included in this TOP are testing and data-reduction procedures for the following subtests: laboratory electronic, field operation, adverse condition, transportability, reliability, maintenance, safety, and human factors.

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

AMSTE-RP-702-105
*Test Operations Procedure 6-2-034 10 April 1989
AD No. A206827

CHRONOGRAPH, FIELD ARTILLERY

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1. SCOPE. This TOP describes tests for a type of Doppler radar system referred to as a radar chronograph. Radar chronographs, such as the M90, are used by the U.S. Army for measuring muzzle velocity in field artillery applications. Included in this TOP are testing and data-reduction procedures for the following subtests: laboratory electronic, field operation, adverse condition, transportability, reliability, maintenance, safety, and human factors.

2. FACILITIES AND INSTRUMENTATION.

2. Facilities.

ITEM	REQUIREMENTS
Weapons and ammunition	As required
Environmental test chambers	See MIL-STD-810D[1]**
Laboratory electronic testing facilities	
Shock test facilities	Shock machines/testers, test fixtures, transducers, oscilloscope-cameras, or multi-band recorders, X-ray inspection equipment, arresting media (see MTP 5-2-506[2])

*Supersedes MTP 6-2-034 dated 17 September 1968.

**Footnote numbers/letters correspond to those in Appendix C.

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3. REQUIRED TEST CONDITIONS.

3.1 Reference Material. The following materials are recommended for review before testing begins:

a. Safety Assessment Report (SAR). This document is required by AR 385-16[6] and should contain any hazards identified by the developer or contractor.

b. Materiel Safety Data Sheet (MSDS).

c. Technical manuals describing the function, operation, and specifications of the test item.

d. Operating instructions and standing operating procedures (SOPs) for all instrumentation used.

e. Reports of previous Doppler system tests and related tests.

f. Inspection checklists that cover common types of unsafe conditions are contained in Appendix C of TOP 1-1-060.[7] Also check other system-peculiar TOPs. When checklists as described above do not adequately cover the test item, appropriate checklists can be developed as described in Appendix C of TOP 1-1-060.

g. All applicable TOPs. The following TOPs provide some pertinent background information on general radar characteristics:

TOP No.	Title
4-1-005[a]	The Doppler Velocimeter
6-2-020[8]	Radar Antenna Tests
6-2-220[b]	Radar Field Artillery
6-2-222[c]	Radar, Target and Ranging
6-2-558[d]	R.F. Power Output
7-3-524[9]	Radar Reflectivity

3.2 Inspection. Upon receipt, the following procedure will be used to inspect the test item:

a. Photograph the packaged test item to record the "as received" condition.

b. Measure and record the size and weight of the packaged test item including cable lengths.

c. Unpack and inspect the test item and its components. Record the following:

(1) Nomenclature

(2) Model number

(3) Serial number

(4) Manufacturer

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4.1.2 Transmitter Test. This is an electronic evaluation of the test item's transmitter characteristics.

a. Connect a directional coupler to the waveguide between the transmitter and antenna system.

b. Perform transmitter tests described in MTP 5-2-530.[11]

NOTE: Since instrumentation that operates on the Doppler principle requires extremely stable signal generators, particular attention should be given to the frequency accuracy and stability test.

4.1.3 Receiver Test. This is an electronic evaluation of the test item's receiver characteristics.

a. Connect a directional coupler to the waveguide between the receiver and the antenna system.

b. Subject the radar chronograph's receiver portion to the applicable tests of MTP 5-2-529.[12]

NOTES:

1. Particular attention should be given to determining the receiver's frequency response and sensitivity, automatic frequency control (AFC) operation, automatic gain control (AGC) operation, and the velocity filter bandwidths, as required.

2. The receiver sensitivity test determines the minimum signal to which the set will respond properly.

3. The Doppler frequency response may be sampled during the receiver sensitivity test, but the test frequencies sampled must be selected carefully since one or more circuits may not respond equally to all required frequencies.

4.1.4 Antenna Test. This is an electronic evaluation of the test item's wave guide-antenna characteristics.

a. Connect a directional coupler to the waveguide between the antenna system and the transmitter/receiver.

b. Subject the radar chronograph's wave guide-antenna system to the applicable tests of TOP 6-2-020 including measurements of the voltage standing wave ratio (VSWR).

4.1.5 Electromagnetic Compatibility. This test determines the electrical compatibility of the module portions (i.e., transmitter, receiver, antenna) of the unit test item and the unit's compatibility with applicable exterior equipment.

While monitoring the applicable measurable chronograph characteristics of paragraphs 4.1.2, 4.1.3, and 4.1.4, perform the following:

a. Subject the test item, as a unit, to the applicable tests of TOP 6-2-560[13] to determine the electrical compatibility of the transmitter-receiver-antenna combination.

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Conduct a comprehensive firing program using various weapon charge combinations and a variety of projectile shapes and sizes. A wide range of muzzle velocities and radar cross-sections is desired.

4.2.4 Performance Tests. These determine the range of velocities and the maximum distance at which the test item can make acceptable measurements.

a. Fire a variety of projectile-charge combinations so that the test item is required to measure extremely low and high projectile velocities.

b. Fire projectiles at a fixed elevation. Re-aim the test item after each round so that the radar beam intersects the trajectory at a farther range. Repeat this until the maximum range of the test item is obtained.

4.3 Adverse Condition Tests.

4.3.1 Environmental Tests. These tests determine whether system performance is degraded when the test item is subjected to particular simulated environments.

Conduct the following tests in accordance with MIL-STD-810D:

a. High temperature tests (Method 501.2, Procedures I and II)

b. Low temperature tests (Method 502.2, Procedures I and II)

c. Humidity tests (Method 507.2, Procedure I or III)

d. Fungus tests (Method 508.3)

e. Salt fog tests (Method 509.2)

f. Rain tests (Method 506.2, Procedure I)

4.4 Transportability Test. This determines transport effects on the test item's physical and operational characteristics.

Determine the transportability of the test item in accordance with TOP 1-2-500. The test documentation, test sponsor, or item user should provide information about the test item's operational environment. Once the operational environment is known, the tests described in TOP 1-2-500 can be tailored accordingly.

4.5 Reliability Evaluation. This test determines if the test item meets reliability specifications.

Use MTP 6-2-503[15] for general guidance in tailoring the test to the test item requirements. (At USACSTA, the Systems Analysis Branch can provide input for the number of measurements required, based on the number of test items and test criteria.) The possibility of combining the firing portion of this evaluation with ongoing ammunition tests should be considered. This will alleviate the necessity of obtaining rounds for test and will also efficiently employ support instrumentation.

4.6 Maintainability. This test evaluates the test item's maintainability characteristics. It will identify maintenance requirements, describe the ease of performing maintenance actions, and identify any special tool requirements.

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5.1.5 Electromagnetic Compatibility. Record all data regarding compatibility/incompatibility with exterior equipment. Perform an equipment search as described in TOP 6-2-560 to determine other electrical equipment which operates at the frequencies measured above.

5.1.6 Power Requirements. Record the data according to MTP 6-2-514.

5.2 Field Operation Tests.

5.2.1 Alignment, Installation, and Operational Checkout.

a. Weapon type

b. Radar chronograph type

c. Outstanding terrain features

d. Alignment procedure for test item

e. Type and model number of power supply

f. Any failures during checkout

g. Any internal adjustments

5.2.2 Trial Firing. Record the following on a round-by-round basis:

a. Firing weapon identification

b. Projectile characteristics:

(1) Type

(2) Description of RF reflective characteristics and/or quantitive data, if available.

(3) Charge identification and weight.

(4) Expected muzzle velocity

c. Test item characteristics:

(1) Observations of internal meter readings.

(2) Projectile velocity versus time data from test item's readout.

(3) Any noticeable shock or blast damage.

5.2.3 Accuracy and Imprecision Test. Record the following for each round fired:

a. Firing weapon identification

b. Projectile characteristics:

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e. Any noticeable shock or blast damage to the test item

f. Any failure of the radar chronograph to determine velocity of a projectile

g. Any component failures encountered during testing

5.3 Adverse Condition Tests.

5.3.1 Environmental Tests.

a. Type of environmental test (humidity, fungus, etc.)

b. Data collected in accordance with MIL-STD-810D.

c. For each firing trial, record the following:

(1) Firing weapon identification

(2) Projectile characteristics:

(a) Type

(b) Description of RF reflective characteristics and/or quantitative data, if available

(c) Charge identification and weight

(3) Test item characteristics:

(a) Observations of internal meter readings

(b) Projectile velocity versus time data from test item output data

(c) Any noticeable shock or blast damage

(4) Calculated and measured values from the velocity measurement instrumentation

(5) Meteorological data:

(a) Air density (kg/m^3)

(b) Air temperature (°C)

(c) Barometric pressure (millibars)

(d) Relative humidity (%)

(e) Wind direction and speed at time of fire in m/s; to be useful, this will require multiple wind sensors elevated so that they are as close as possible to the bullet's flight path

(f) Test site altitude

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5.8 Human Factors Evaluation. Record data according to TOP 1-2-610.

6. PRESENTATION OF DATA.

a. Failure of the test item to meet specified electrical and safety criteria shall be recorded and possible causes determined. Safety hazards shall be categorized according to MIL-STD-882B[19] and TOP 1-1-012.[20] These findings will be a combination of observations taken throughout all phases of testing.

b. Data obtained from the standard velocity check instrumentation shall be reduced as described in TOP 4-2-805 to obtain the projectile's velocity versus time history. The DDTST already incorporates the software required to perform this computation and can provide this information in near real time. If sky screens are used, ballistic data for the rounds fired will be required. Accumulate the required data and compute the velocity versus time history. Compare the velocities from the standards to the test item velocities to yield a difference between readings. The imprecision can then be estimated using the procedure in Appendix A. Tabulate the imprecision data as a function of range, projectile, charge, weapon, etc. Range information for each reading can be obtained by integrating the velocity versus time information to yield range versus time and then solving for the time of each estimate. The DDTST, again, is capable of this computation for the reference data. If the test item does not have gross errors in velocity measurement, solving the DDTST's range versus time equations will give an adequate estimate of the projectile range at the time of measurement.

c. Descriptive evaluations of the test item's performance under adverse conditions will be provided. These evaluations shall document any performance degradation observed. This shall be presented as a percent error from a standard. For the case of muzzle velocity, this would be

$$\% \text{ error} = \frac{MVT-MVS}{MVS} * 100$$

in which

MVS = muzzle velocity from the standard
MVT = muzzle velocity from the test item

Also, any degradation from the percent error in the accuracy portion of the test should be reported.

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APPENDIX A

STATISTICAL PROCEDURES

The estimate of imprecision for the reference items, when assuming data are from the same population, may be written:

$$\sigma_A = \sigma_B = S_{A-B} / \sqrt{2}$$

in which

σ_A = estimate of imprecision for reference set A.
σ_B = estimate of imprecision for reference set B.
S_{A-B} = sample standard deviation of the difference between set A and set B.

The estimate of average level for the reference items may be written:

$$\overline{V_R} = 1/2 \ (\overline{V_A} + \overline{V_B})$$

The estimate of imprecision for the test item may be written:

$$\sigma_T = \left[S^2_{T-0.5(A+B)} - 0.25 \ S^2_{A-B} \right]^{1/2}$$

in which

$S^2_{T-0.5(A+B)}$ = sample variance of the difference between the test item and the average of the reference (set A and set B).

$S^2_{(A-B)}$ = sample variance of the difference between the reference (set A and set B).

The average level of the test is assessed based on the difference between the average level of the test and the average level of reference expressed as a percent of the average level of the reference. These tests should be applied in accordance with reference i.

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APPENDIX B

HAZARD ANALYSIS

1. Can someone be struck by some moving object in the system?

   Look for: -normally moving objects
   -normally stationary objects.

2. Can someone strike against some object?

   Look for: -protruding objects
   -permanent objects requiring an effort to avoid
   -cramped work areas
   -manipulations that require great effort.

3. Can someone be caught between two objects?

   Look for: -a normally moving object approaching a stationary one
   -a normally stationary object forced to move
   -two normally moving objects contacting

4. Can someone be contacted by a substance that can cause injury?

   Look for: -hot or cold surfaces
   -electrically charged equipment
   -corrosive chemicals
   -exhaust fumes.

5. Can someone be exposed to a harmful condition?

   Look for: -toxic gases, fumes or vapors
   -air-borne particles
   -extreme heat or cold
   -oxygen-deficient atmosphere
   -radioactive radiation
   -intense light.
   -high noise level
   -blast overpressure

6. Can someone slip, trip or fall?

7. Can someone be over-exerted?

   Look for: -manual handling of heavy objects
   -imbalanced equipment.

8. Can someone be caught in/on something?

   Look for: -single-entry enclosures
   -small exposed floor openings
   -tight or cramped places
   -stationary or moving projecting objects.

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APPENDIX C

REFERENCES

Required References

1. MIL-STD-810D, Environmental Test Methods and Engineering Guidelines, 19 July 1983; Notice 1, 31 Jul 86.

2. MTP 5-2-506, Shock Test Procedures, December 1966.

3. MTP 5-2-507, Vibration Test, 10 April 1967.

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5. TOP 1-2-500, Transportability, 7 February 1973; Change 1, 22 July 1976; Change 2, 24 August 1976; Change 3, 20 March 1979.

6. AR 385-16, System Safety Engineering and Management, 3 September 1985; AR Suppl 1, 1 Dec 86; AMC Suppl 1, 22 Jan 82; TECOM Suppl 1, 11 Jun 82; and APG Suppl 1, 5 Jan 83.

7. TOP 1-1-060, System Safety Engineering, 7 April 1986.

8. ITOP 6-2-020, Radar Antenna Tests, 15 April 1988.

9. TOP 7-3-524, Radar Reflectivity, 10 June 1986.

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11. MTP 5-2-530, Transmitter Tests, 5 December 1967.

12. MTP 5-2-529, Radar Receivers, 7 December 1967.

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14. TOP 6-2-514, Electrical Power Requirements, 15 October 1981.

15. MTP 6-2-503, Reliability, 23 March 1970.

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17. MTP 3-2-616, Radio Frequency Radiation Hazards to Personnel, 12 June 1968.

18. TOP 1-2-610, Human Factors Engineering, Part I, Test Procedures; Part II, Human Factors Engineering Data Guide for Evaluation (HEDGE), 30 November 1983.

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